

This is the Canonical Logical Data Model for V1. It is the definitive, technology-agnostic specification bridging the domain model and physical database design. It assumes a single-issuer, self-hosted deployment operating entirely through manual UI interactions, supporting exclusively FE (01) and CCF (03) documents, plain invalidation, offline contingency, and strict immutable evidence.

All relationships are conceptual identity references. Physical foreign keys, constraints, and data types are deferred to physical design.

1. Core Value Objects: Money & Snapshots

Before defining aggregates, these foundational structures dictate data integrity.

1.1 Money Model

Purpose: Ensure absolute precision and reconciliation of all financial data without floating-point errors.

Immutability: Strictly immutable once instantiated.

Rounding Behavior: All fractional arithmetic must be rounded to a fixed scale (legally mandated decimal places, typically 2 or 4) *before* aggregation.

- **Money (Value Object):**
 - Amount (Decimal): The numeric value. Must never be negative unless explicitly modeling a discount line (not applicable in V1).
 - Currency (String): Fixed to "USD" for V1.
- **TaxLine (Value Object):**
 - TaxCategory (String): The classification (e.g., Standard IVA 13%, Exempt, Non-Subject).
 - TaxableBase (Money): The amount to which the tax is applied.
 - ComputedTax (Money): The exact tax amount calculated.
- **TaxSummary (Value Object):**
 - TaxLines (List of TaxLine): The aggregated sum of taxes grouped by TaxCategory.
- **DocumentTotals (Value Object):**
 - NetTotal (Money): Sum of all line amounts before tax.
 - TaxTotal (Money): Sum of all ComputedTax from the TaxSummary.
 - GrossTotal (Money): The final payable amount.
 - *Reconciliation Rule:* GrossTotal MUST exactly equal NetTotal + TaxTotal. A document cannot be Validated if this invariant fails.

1.2 Snapshot Model

Purpose: Isolate historical fiscal documents from subsequent changes in master data.

Immutability: Permanently immutable from the moment of capture.

Trigger: Captured atomically during the Draft $\rightarrow$ Validated and Validated $\rightarrow$ Signed transitions.

- **IssuerSnapshot (Value Object):**
 - LegalName (String, Req): Name of the issuer at the time of signing.
 - TaxRegistration (Value Object, Req): NIT, NRC, and Economic Activity Code.
 - Address (Value Object, Req): Department, Municipality, specific location details.

- IssuingPointCoordinates (Value Object, Req): The exact Establishment and POS codes that generated the document.
- **ReceptorSnapshot (Value Object):**
 - TaxIdentity (Value Object, Req): Discriminator (Consumer vs. Taxpayer) and the ID string (DUI, NIT, NRC). FE may have relaxed identity; CCF requires full taxpayer identity.
 - LegalName (String, Req): Name of the counterparty.
 - Address (Value Object, Opt for FE/Req for CCF): Structured location.
- **ItemSnapshot (Value Object):**
 - Description (String, Req): The human-readable product/service name.
 - UnitOfMeasure (String, Req): Catalog-defined unit (e.g., "Unit", "Service", "Hour").
 - TaxBehavior (String, Req): The applied tax rule (Standard, Exempt).

2. Issuer Configuration Model

Purpose: The central definition of the singleton legal entity running the software.

Ownership: Configuration Context.

Lifecycle: Initialized at deployment. Modifiable only by users with the Owner role.

Invariants: Exactly one active configuration exists system-wide.

2.1 IssuerConfiguration (Aggregate Root)

- **Identity Strategy:** Singleton (Global ID).
- **Attributes:**
 - ConfigurationId (Identifier, Req): Internal ID. (Immutable)
 - LegalName (String, Req): Official business name. (Mutable)
 - TaxRegistrationData (Value Object, Req): NIT, NRC, Economic Activity. (Mutable)
 - Address (Value Object, Req): Headquarters address. (Mutable)
 - CertificateReference (Value Object, Req): Pointer/URI to the secure keystore location of the cryptographic signing key. Never contains raw key material. (Mutable)
 - AuthorityCredentialsReference (Value Object, Req): Pointer/URI to the secure vault containing tax authority API tokens/passwords. (Mutable)
 - Establishments (List of Entities, Req): Minimum of one. (Mutable)

2.2 Establishment (Entity)

- **Identity:** EstablishmentCode (String, unique within Issuer).
- **Parent Aggregate:** IssuerConfiguration.
- **Attributes:**
 - Name (String, Req): Descriptive name. (Mutable)
 - Address (Value Object, Req): Physical location. (Mutable)
 - PointsOfSale (List of Entities, Req): Minimum of one. (Mutable)

2.3 PointOfSale (Entity)

- **Identity:** POSCode (String, unique within Establishment).
- **Parent Aggregate:** IssuerConfiguration.
- **Attributes:**

- Description (String, Req): Logical or physical name. (Mutable)
- IsActive (Boolean, Req): Soft-delete toggle. (Mutable)

3. Issuance Context

3.1 FiscalDocument (Aggregate Root)

Purpose: The central transactional record representing a single DTE from composition to terminal state.

Ownership: Issuance Context.

Lifecycle: Draft $\rightarrow$ Validated $\rightarrow$ Signed $\rightarrow$ (Submitted | Contingency-Held) $\rightarrow$ (Accepted | Rejected | Invalidated).

Invariants:

1. Identity (Generation Code + Control Number) is assigned exactly once and never regenerated.
2. DTE Type is fixed at creation.
3. Cannot transition out of Draft unless DocumentTotals perfectly reconcile.
4. Entirely immutable once Status reaches Signed.

Attributes:

- DocumentId (Identifier, Req): Internal database identity. (Immutable)
- DTEType (String, Req): '01' or '03'. Bound at creation. (Immutable)
- SchemaVersion (String, Opt): Pinned version from Registry. Populated at Validation. (Immutable once set)
- GenerationCode (UUID-String, Opt): Globally unique MH identifier. Populated at Validation. (Immutable once set)
- ControlNumber (String, Opt): Formatted correlative (e.g., DTE-01-M001P001-0000000000000001). Populated at Validation. (Immutable once set)
- Status (Enum, Req): Current state. (Mutable via state machine)
- OperationCondition (Enum, Req): Cash or Credit. (Mutable in Draft)
- IssuerSnapshot (IssuerSnapshot, Opt): Populated at Signing. (Immutable once set)
- ReceptorSnapshot (ReceptorSnapshot, Opt): Populated at Signing. (Immutable once set)
- Lines (List of DocumentLine, Req): Must contain $\geq$ 1 line. (Mutable in Draft)
- TaxSummary (TaxSummary, Req): Computed aggregation. (Mutable in Draft)
- Totals (DocumentTotals, Req): Reconciled amounts. (Mutable in Draft)
- ContingencyEpisodeId (Identifier, Opt): Reference to an episode if issued offline. (Immutable once set)

3.2 DocumentLine (Entity)

Purpose: Represents a single billed good/service inside the document.

Parent Aggregate: FiscalDocument.

Identity: LineNumber (Integer, strictly sequential within the document).

Lifecycle: Created/Edited in Draft; Frozen at Signing.

Attributes:

- ItemSnapshot (ItemSnapshot, Req): Frozen view of the catalog item or manual entry.

- Quantity (Decimal, Req): Must be $\geq 0\$$.
- UnitPrice (Money, Req): Must be $\geq 0\$$.
- TaxLines (List of TaxLine, Req): Taxes applicable strictly to this line.
- LineTotal (Money, Req): $\text{Quantity} * \text{UnitPrice} + \text{applicable taxes (if gross)}$.

3.3 ControlNumberSeries (Aggregate Root)

Purpose: Single authority for the gapless, ordered correlatives.

Ownership: Issuance Context.

Lifecycle: Created on system configuration; strictly increments.

Invariants: Assignments are atomic. Monotonic and gapless. Correlatives are never returned or rewound.

Attributes:

- SeriesId (Identifier, Req): Internal identity. (Immutable)
- DTEType (String, Req): e.g., '01'. (Immutable)
- EstablishmentCode (String, Req): Scope. (Immutable)
- POSCode (String, Req): Scope. (Immutable)
- NextCorrelative (Integer, Req): The integer ready to be allocated. (Mutable strictly via atomic increment)
- *Business Constraint:* Uniqueness guaranteed on (DTEType, EstablishmentCode, POSCode).

3.4 TransmissionRecord (Aggregate Root)

Purpose: The append-only, exhaustively complete truth of every interaction with the authority regarding a specific payload.

Ownership: Issuance Context.

Lifecycle: Opened on first transmit attempt; Closed when outcome is Terminal.

Invariants: Never edited. Never deleted. Outcomes are derived strictly from authority responses, never inferred.

Attributes:

- RecordId (Identifier, Req): Internal identity. (Immutable)
- TargetIdentity (Identifier, Req): Pointer to the FiscalDocument or InvalidationEvent or ContingencyNotification being transmitted. (Immutable)
- TargetType (Enum, Req): "Document" | "Invalidation" | "ContingencyNotification". (Immutable)
- LifecycleStatus (Enum, Req): Open, Terminal_Accepted, Terminal_Rejected, Exhausted. (Mutable)
- Attempts (List of TransmissionAttempt, Req): Ordered history. (Append-only)

Entity: TransmissionAttempt

- **Identity:** AttemptSequence (Integer, monotonic within record).
- **Attributes:**
 - AttemptTimestamp (DateTime, Req): Exact time of transmission. (Immutable)
 - RequestPayload (Blob/String, Req): The exact byte structure sent. (Immutable)
 - ResponsePayload (Blob/String, Opt): The exact byte structure received (null if network

- timeout). (Immutable)
- NetworkError (String, Opt): Exception details if the authority was unreachable. (Immutable)
- InterpretedOutcome (Enum, Req): Success, Rejected, ContingencySignal, Timeout, AuthorityError. (Immutable)
- Observations (List of Strings, Opt): Descriptive error codes/messages parsed from the authority. (Immutable)

5. Invalidation Context

5.1 InvalidationEvent (Aggregate Root)

Purpose: Execute the lawful voiding of a single accepted document (Plain Invalidation).

Ownership: Invalidation Context.

Lifecycle: Created $\rightarrow$ Submitted $\rightarrow$ (Accepted | Rejected).

Invariants: Target document MUST be in Accepted state. Only one Accepted invalidation can exist per document.

Attributes:

- InvalidationId (Identifier, Req): Internal identity. (Immutable)
- TargetDocumentId (Identifier, Req): Reference to the original Fiscal Document. (Immutable)
- ReasonType (Enum, Req): Limited in V1 to "Rescind" (Type 2) or "Other" (Type 3). (Immutable)
- Justification (String, Req): Human-provided rationale. (Immutable)
- TransmissionRecordId (Identifier, Opt): Pointer to its transmission lifecycle. (Immutable once set)
- Status (Enum, Req): Draft, Submitted, Accepted, Rejected. (Mutable via state machine)

6. Contingency Context

6.1 ContingencyEpisode (Aggregate Root)

Purpose: Govern bounded periods of lawful offline issuance.

Ownership: Contingency Context.

Lifecycle: Open $\rightarrow$ Reconciling $\rightarrow$ Closed.

Invariants: Only one episode can be Open at a time. Cannot be closed until Notification is accepted and all held documents are terminal.

Attributes:

- EpisodeId (Identifier, Req): Internal identity. (Immutable)
- OpenedAt (DateTime, Req): Start time. (Immutable)
- ClosedAt (DateTime, Opt): End time. (Mutable)
- ReasonCode (String, Req): Authority-defined contingency cause (e.g., "MH Unavailable"). (Immutable)
- Status (Enum, Req): Open, Reconciling, Closed, WindowBreached. (Mutable)
- NotificationTransmissionId (Identifier, Opt): Pointer to the record tracking the required MH

notification event. (Mutable)

- EnrolledDocuments (List of Identifiers, Req): Pointers to the FiscalDocuments held under this episode. (Append-only while Open)

7. Parties & Catalog Context

7.1 Counterparty (Aggregate Root)

Purpose: Reusable receptor master data.

Ownership: Parties Context.

Lifecycle: Created $\rightarrow$ Edited $\rightarrow$ Deactivated.

Invariants: Tax identity validation rules apply (CCF requires strict checks). Editing this record NEVER alters a ReceptorSnapshot on an existing document.

Attributes:

- CounterpartyId (Identifier, Req): Internal identity. (Immutable)
- TaxpayerKind (Enum, Req): FinalConsumer, Contribuyente. (Mutable)
- TaxIdentityString (String, Opt): The DUI, NIT, or NRC. (Mutable)
- LegalName (String, Req): (Mutable)
- Address (Value Object, Opt): (Mutable)
- IsActive (Boolean, Req): Soft-delete flag. (Mutable)

7.2 Item (Aggregate Root)

Purpose: Reusable product master data.

Ownership: Catalog Context.

Lifecycle: Created $\rightarrow$ Edited $\rightarrow$ Deactivated.

Invariants: Edits NEVER alter an ItemSnapshot on existing document lines.

Attributes:

- ItemId (Identifier, Req): Internal identity. (Immutable)
- InternalSKU (String, Opt): Custom catalog code. (Mutable)
- Description (String, Req): (Mutable)
- DefaultUnitPrice (Money, Req): Must be ≥ 0 \$. (Mutable)
- UnitOfMeasureCode (String, Req): Authority-mapped unit. (Mutable)
- DefaultTaxBehavior (String, Req): (Mutable)
- IsActive (Boolean, Req): Soft-delete flag. (Mutable)

8. Evidence Context

8.1 EvidenceBundle (Aggregate Root)

Purpose: Immutable, append-only proof vault for exactly one FiscalDocument.

Ownership: Evidence Context.

Lifecycle: Created when document is Signed; never closed, never deleted.

Invariants: Cannot modify or delete artifacts. Retained for the full legal horizon regardless of document status (including Rejected).

Attributes:

- BundleId (Identifier, Req): Internal identity. (Immutable)
- DocumentId (Identifier, Req): Pointer to the Fiscal Document. (Immutable)
- Artifacts (List of EvidenceArtifact, Req): (Append-only)

8.2 EvidenceArtifact (Entity)

Purpose: A single incontrovertible proof item.

Parent Aggregate: EvidenceBundle.

Identity: ArtifactId (Identifier, unique system-wide).

Trigger: Generated automatically via domain events (DocumentSigned, DocumentAccepted, InvalidationAccepted).

Attributes:

- ArtifactType (Enum, Req): SignedPayload, AuthorityResponse, AcceptanceStamp, HumanReadablePDF, InvalidationPayload. (Immutable)
- CapturedAt (DateTime, Req): Timestamp of creation. (Immutable)
- ContentBlob (Blob, Req): The actual byte content (JSON, PDF binary, String). (Immutable)
- ContentHash (String, Req): SHA-256 (or stronger) hash of the ContentBlob calculated instantly upon capture to prove integrity. (Immutable)

9. Audit Context

9.1 AuditTrail (Aggregate Root)

Purpose: Cryptographically verifiable chronological record of actions.

Ownership: Audit Context.

Lifecycle: Begins at initialization; grows indefinitely.

Invariants: Append-only. Tamper-evident hash chain linking sequential records.

Attributes:

- TrailId (Identifier, Req): Global singleton ID. (Immutable)
- Entries (List of AuditEntry, Req): (Append-only)

9.2 AuditEntry (Entity)

Purpose: A single verifiable action.

Parent Aggregate: AuditTrail.

Identity: SequenceNumber (Integer, strictly monotonic).

Attributes:

- Timestamp (DateTime, Req): Exact server time. (Immutable)
- ActorIdentity (Identifier, Req): Pointer to the User. (Immutable)
- ActorRole (String, Req): Role at the time of the action. (Immutable)
- ActionType (String, Req): e.g., "DOCUMENT_SIGNED", "EPISODE_OPENED", "CONFIG_CHANGED". (Immutable)
- TargetIdentity (Identifier, Req): ID of the subject aggregate. (Immutable)
- OutcomeStatus (String, Req): Success, Failure, Denied. (Immutable)
- PreviousHash (String, Req): The CurrentHash of the immediately preceding AuditEntry.

Genesis block contains a known seed. (Immutable)

- CurrentHash (String, Req): Cryptographic hash of [PreviousHash + Timestamp + ActorIdentity + ActionType + TargetIdentity + OutcomeStatus]. (Immutable)

10. Access & Registry Contexts

10.1 User (Aggregate Root)

Purpose: Represents a human actor.

Ownership: Access Context.

Lifecycle: Created $\rightarrow$ Role Assigned $\rightarrow$ Deactivated.

Invariants: Must have exactly one role. System must retain at least one active Owner.

Attributes:

- UserId (Identifier, Req): Internal identity. (Immutable)
- Email (String, Req): Login credential. (Mutable)
- Role (Enum, Req): Owner or Biller. (Mutable by Owner)
- IsActive (Boolean, Req): Soft-delete. (Mutable)

10.2 DocumentTypeSpecification (Aggregate Root)

Purpose: Fixed rule set defining structure and behavior per DTE Type behind a seam.

Ownership: Registry Context.

Lifecycle: Fixed/Read-only in V1.

Attributes:

- TypeCode (String, Req): Identity ('01', '03'). (Immutable)
- SchemaVersionBinding (String, Req): The pinned MH version (e.g., 'v1.0'). (Immutable)
- RequiresTaxpayerReceptor (Boolean, Req): True for 03, False for 01. (Immutable)
- AllowedTaxBehaviors (List of Strings, Req): Valid codes for this type. (Immutable)